Specifications for the Description of Rice Germplasm Resources

Author-prepared English reference translation

This reference translation follows the Chinese crop descriptor specification used for the CropMaestro study. It is not an official English edition issued by the source platform. Original section numbers and category-code mappings are retained.

1 Scope

These specifications define the descriptors for rice germplasm resources and their grading criteria.

These specifications apply to the collection, curation and conservation of rice germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative references

The provisions of the following documents become provisions of these specifications through reference herein. For dated references, subsequent amendments (excluding corrections of errors) or revised editions do not apply to these specifications. However, parties entering into agreements based on these specifications are encouraged to consider whether the latest editions of these documents may be used. For undated references, the latest edition applies.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659 Codes for the Names of Countries and Regions of the World

GB/T 2260 Codes for the Administrative Divisions of the People's Republic of China

GB/T 12404 Codes for Institutional Affiliation

GB/T 3543.6-1995 Rules for Agricultural Seed Testing - Determination of Moisture Content

GB/T 5495-1985 Inspection of Grain and Oilseeds - Method for Determining the Brown Rice Recovery of Paddy

GB/T 5492-1985 Inspection of Grain and Oilseeds - Methods for the Identification of Colour, Odour and Taste

GB/T 17891-1999 High-Quality Paddy

GB /T 15683-1995 Determination of Amylose Content in Rice

GB/T 2905-1982 Method for Determining Crude Protein in Seeds of Cereal and Leguminous Crops

GB/T 5519-1988 Method for Determining Thousand-Grain Weight of Grain and Oilseeds

GB/T 4801-1984 Method for Determining Lysine in Cereal Grains

GB/T 2906-1982 Method for Determining Crude Fat in Seeds of Cereal and Oilseed Crops

NY/T 11-1985 Method for Determining Crude Starch in Cereal Grains

NY/T 83-1988 Methods for Determining Rice Quality

NY/T-593-2002 Quality of Rice Varieties for Food Use

NY/T 594-2002 Japonica Rice for Food Use

NY/T 595-2002 Indica Rice for Food Use

GB/T 195557.7-2004 Guidelines for the Conduct of Tests for Distinctness, Uniformity and Stability of New Plant Varieties - Rice

Standard Evaluation System for Rice (SES), IRRI, 2002

GB/T 15790-1995 Rules for the Investigation and Forecasting of Rice Blast

GB/T 15791-1995 Rules for the Investigation and Forecasting of Rice Sheath Blight

GB/T 15792-1995 Rules for the Investigation and Forecasting of the Striped Rice Stem Borer

GB/T 15794-1995 Rules for the Investigation and Forecasting of Rice Planthoppers

3 Terms and definitions

3.1 Rice

Rice belongs to the family Gramineae and the genus Oryza. The genus Oryza comprises two cultivated species, Asian cultivated rice (Oryza sativa L.) and African cultivated rice (Oryza glaberrima Steud.), and 20-25 wild species. Rice (Oryza sativa L.) has a chromosome number of 2n=24.

3.2 Rice germplasm resources

Wild rice resources, landraces, bred varieties, breeding lines, hybrid rice resources, genetic stocks, mutants and other materials.

3.3 Basic information

Information describing the basic attributes of rice germplasm resources, including the national unified identifier, germplasm name, scientific name, origin and germplasm type.

3.4 Morphological and biological characteristics

Characteristics of rice germplasm resources such as phenological stages, botanical morphology and yield traits.

3.5 Quality characteristics

Milling quality, appearance quality, cooking and eating quality, nutritional quality and other quality characteristics of rice germplasm resources. These include brown rice recovery, milled rice recovery, head rice recovery, brown rice shape, percentage of chalky kernels, chalky area size, degree of chalkiness, translucency, gelatinization temperature, gel consistency, amylose content, protein content, lysine content, fat content and aroma.

3.6 Abiotic stress resistance

The ability of rice germplasm resources to adapt to or withstand abiotic stresses, including drought resistance, salt tolerance, alkali tolerance, cold tolerance and heat tolerance.

3.7 Resistance to diseases and insect pests

The ability of rice germplasm resources to adapt to or withstand biotic stresses. These include resistance to rice blast, bacterial blight, sheath blight, bacterial leaf streak, whitebacked planthopper, brown planthopper, rice gall midge, striped stem borer and yellow stem borer.

4 Basic information

4.1 National unified identifier

The unique identifier for a germplasm resource. National unified identifiers are assigned according to variety type and origin. For example, identifiers for bred varieties begin with ZD; Beijing landraces with 01; introduced varieties with WD; hybrid rice combinations with H; male sterile lines with A; maintainer lines with B; and restorer lines with R.

4.2 Genebank accession number

The accession number of the rice germplasm in the National Long-Term Crop Germplasm Bank.

4.3 Introduction number

The number assigned to rice germplasm when it is introduced from abroad.

4.4 Collecting number

The number assigned to rice germplasm when it is collected in the wild.

4.5 Germplasm name

The Chinese name of the rice germplasm.

4.6 Foreign-language germplasm name

The foreign-language name of germplasm introduced from abroad, or the name in Hanyu Pinyin of domestic germplasm.

4.7 Family name

Gramineae (the grass family).

4.8 Genus name

Oryza (the rice genus).

4.9 Scientific name

The scientific name of rice (Asian cultivated rice) is Oryza sativa L.

4.10 Country of origin

The name of the country, region or international organization of origin of the rice germplasm.

4.11 Province of origin

The province of origin of the rice germplasm.

4.12 Locality of origin

The names of the county, township and village of origin of the rice germplasm.

4.13 Elevation

The elevation of the place of origin of the rice germplasm. Unit: m.

4.14 Longitude

The longitude of the place of origin of the rice germplasm, expressed in degrees (º) and minutes (′), in the format DDDFF, where DDD denotes degrees and FF denotes minutes.

4.15 Latitude

The latitude of the place of origin of the rice germplasm, expressed in degrees (º) and minutes (′), in the format DDFF, where DD denotes degrees and FF denotes minutes.

4.16 Source locality

For rice germplasm introduced from abroad, the name of the country, region or international organization from which it was obtained; for domestic germplasm, the names of the province and county from which it was obtained.

4.17 Original holding institution

The name of the institution that held the rice germplasm before its submission to the National Long-Term Crop Germplasm Bank.

4.18 Accession number at the original holding institution

The accession number assigned to the rice germplasm by the original holding institution.

4.19 Pedigree

The parentage and genetic relationships of bred rice varieties or lines.

4.20 Breeder

The name of the institution or individual that bred the rice variety or line.

4.21 Year of development

The year in which the rice variety or line was successfully developed.

4.22 Breeding method

The method used to breed the rice variety or line, including pure-line selection, hybridization breeding, mutation breeding, ploidy breeding, molecular breeding and other methods.

4.23 Germplasm type

Rice germplasm is classified into 8 types.

1	Wild resources
2	Landraces
3	Bred varieties
4	Breeding lines
5	Hybrid rice resources
6	Genetic stocks
7	Mutants
8	Other

4.24 Image

The filename of the image of the rice germplasm. The image format is .jpg.

4.25 Observation site

The name of the site where the morphological and biological characteristics of the rice germplasm are observed.

5 Morphological and biological characteristics

5.1 Subspecies type

Rice is classified into 2 subspecies, indica and japonica.

1	Indica
2	Japonica

5.2 Lowland or upland type

Rice is classified into lowland rice and upland (dryland) rice.

1	Lowland rice
2	Upland (dryland) rice

5.3 Endosperm type

The non-glutinous or glutinous nature of the endosperm.

1	Non-glutinous rice
2	Glutinous rice

5.4 Photoperiod and temperature response

Differences in sensitivity to photoperiod and temperature.

1	Early-season rice
2	Mid-season rice
3	Late-season rice

5.5 Maturity class

The overall expression of maturity characteristics of rice climatic ecotypes.

1	Early-maturing
2	Intermediate-maturing
3	Late-maturing

5.6 Sowing date

The date on which the seeds are sown for the evaluation of the morphological and biological characteristics of the rice germplasm. Expressed as year, month and day in the format YYYYMMDD.

5.7 Initial heading date

The date on which 10% of the rice panicles in the entire plot have headed. An individual panicle is considered to have headed when 3 cm of the panicle has emerged beyond the leaf sheath. Expressed as year, month and day in the format YYYYMMDD.

5.8 Heading date

The date on which 50% of the rice panicles in the entire plot have headed. Expressed as year, month and day in the format YYYYMMDD.

5.9 Full heading date

The date on which 80% of the rice panicles in the entire plot have headed. Expressed as year, month and day in the format YYYYMMDD.

5.10 Maturity date

The date on which 90% or more of the filled grains in the entire plot have turned yellow and ripe. Expressed as year, month and day in the format YYYYMMDD.

5.11 Total growth duration

The number of days from the date of sprouting to the date of maturity. Unit: d.

5.12 Plant height

After rice has matured, the distance from ground level to the tip of the highest panicle on the plant, excluding awns. Unit: cm.

5.13 Culm length

The length from ground level to the panicle neck node of the rice plant. Unit: cm.

5.14 Panicle length

After rice has matured, the distance from the panicle neck node to the panicle tip, excluding awns. Unit: cm.

5.15 Number of spikelets per panicle

The total number of spikelets on each panicle, i.e. the sum of filled grains, unfilled or shrivelled grains, and shed grains. Unit: grains.

5.16 Number of effective panicles

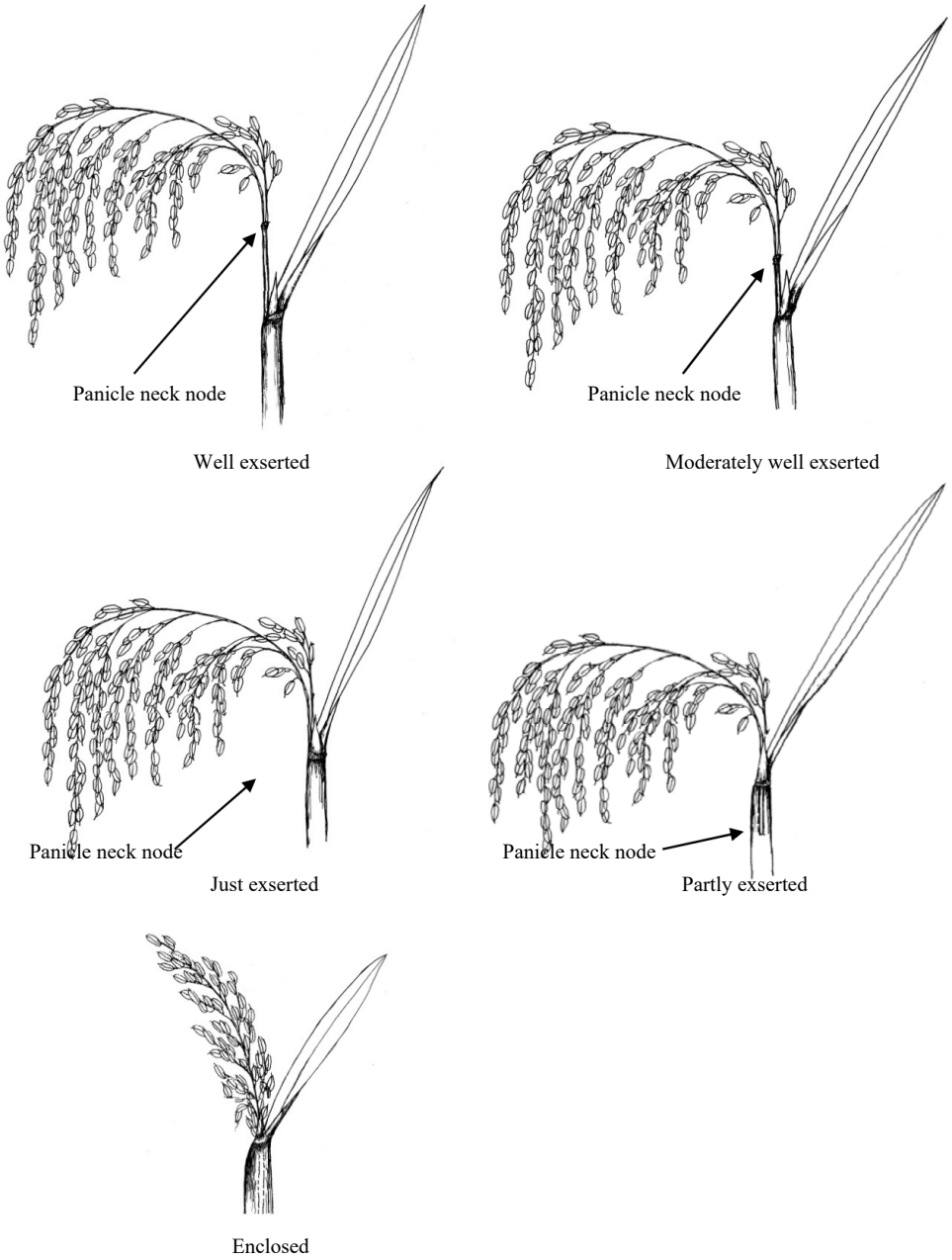
The total number of effective panicles per plant. A panicle that has headed and has at least 5 spikelets, as counted in 5.15, is considered an effective panicle.

5.17 Panicle exsertion

The distance from the flag-leaf collar to the panicle neck node (Figure 1).

1	Well exserted
3	Moderately well exserted
5	Just exserted
7	Partly exserted
9	Enclosed

Figure 1. Panicle exsertion

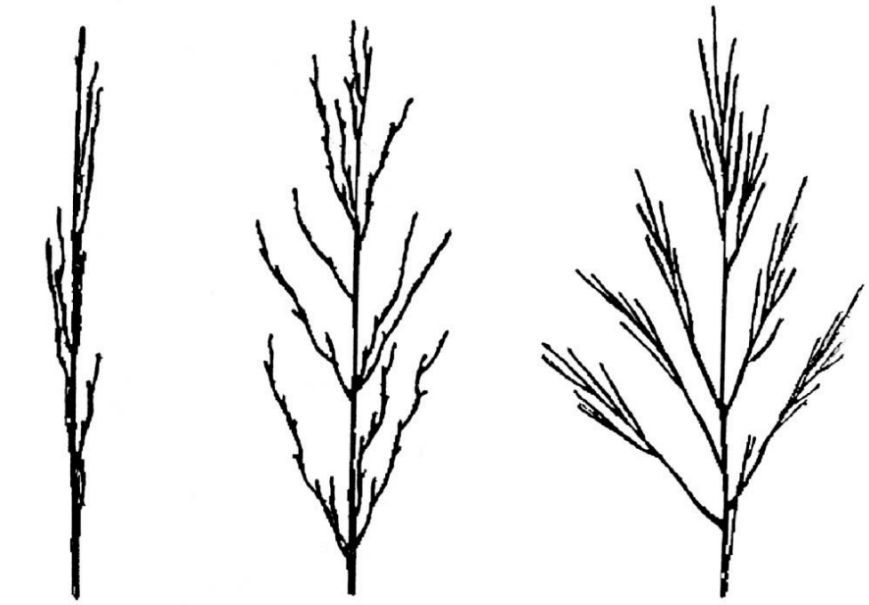


5.18 Panicle type

The degree of dispersion or compactness of the panicle branches (Figure 2).

1	Compact
2	Intermediate
3	Open

Figure 2. Panicle type

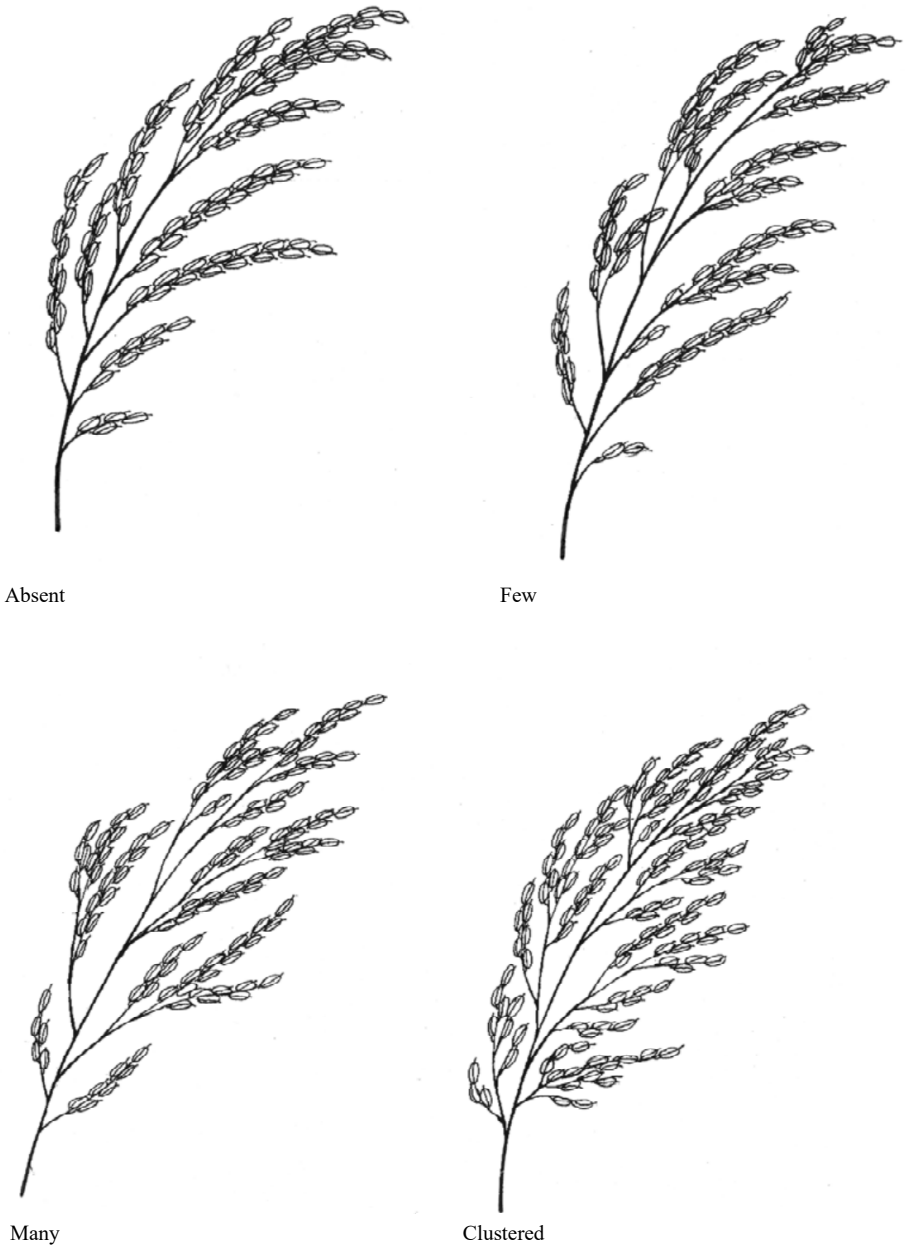


5.19 Secondary branching

The pattern of spikelets borne on the secondary branches of the rice panicle (Figure 3).

1	Absent
5	Few
7	Many
9	Clustered

Figure 3. Spikelets borne on secondary branches



5.20 Panicle attitude

The erectness of the panicle axis, evaluated by the angle between the line connecting the flag-leaf collar to the panicle tip and the extension of the culm (Figure 4).

1	Erect
5	Semi-erect
7	Curved
9	Drooping

Figure 4. Panicle attitude



5.21 Seed-setting rate

The mean number of filled grains as a proportion of the mean total number of spikelets per panicle. Expressed as a percentage (%).

5.22 Thousand-grain weight

The weight of 1000 relatively well-filled paddy grains, as determined by settling after washing in water. Unit: g.

5.23 Paddy grain length

The length from the base of the sterile lemmas at the lowermost part of the grain to the tip of whichever is longer, the palea or lemma (the apiculus). Unit: mm.

5.24 Paddy grain width

The distance across the widest lateral part of the lemma and palea. Unit: mm.

5.25 Paddy grain thickness

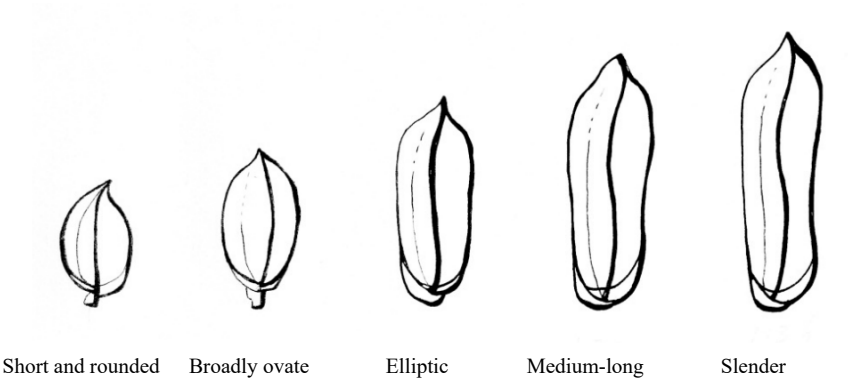
The distance between the lemma and palea at the thickest part of the grain. Unit: mm.

5.26 Paddy grain shape

Evaluated using the ratio of paddy grain length to width (Figure 5).

1	Short and rounded
3	Broadly ovate
5	Elliptic
7	Medium-long
9	Slender

Figure 5. Paddy grain shape



5.27 Brown rice length

The length of the brown rice kernel (the dehulled grain). Unit: mm.

5.28 Brown rice width

The width of the brown rice kernel at its widest point. Unit: mm.

5.29 Brown rice thickness

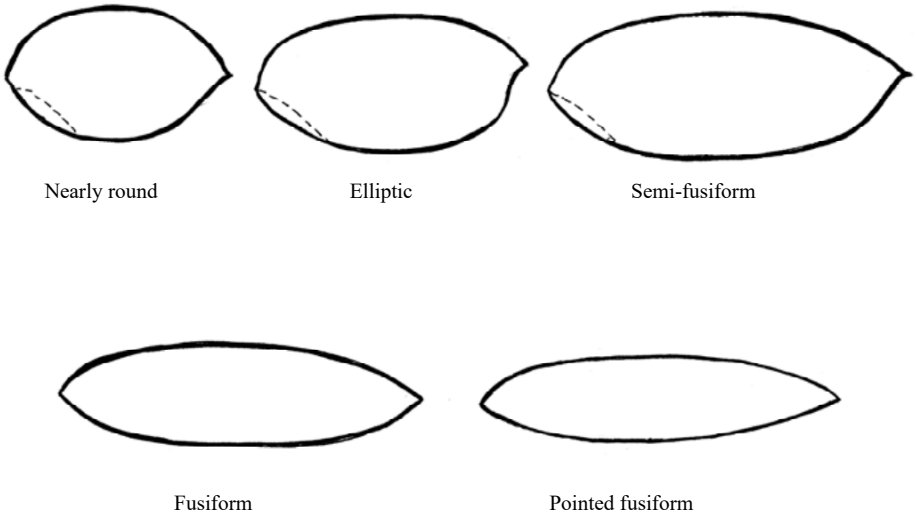
The thickness of the brown rice kernel at its thickest point. Unit: mm.

5.30 Brown rice shape

Evaluated according to the shape of the brown rice kernel (Figure 6).

1	Nearly round
3	Elliptic
5	Semi-fusiform
7	Fusiform
9	Pointed fusiform

Figure 6. Brown rice shape



5.31 Seed coat colour

The colour of the seed coat of the brown rice kernel.

1	White
2	Red
3	Brown
4	Purple
5	Black

5.32 Coleoptile colour

The colour of the coleoptile.

1	Colourless
2	Light purple
3	Dark purple

5.33 Leaf sheath colour

The colour of the leaf sheath.

1	Yellow
2	Green
3	Purple

5.34 Leaf blade colour

The colour of the leaf blade.

1	Light yellow
2	Yellow spots
3	Green-and-white variegation
4	Light green
5	Green
6	Dark green
7	Purple margins
8	Purple spots
9	Purple

5.35 Leaf blade pubescence

The distribution of hairs on the surface of the leaf blade.

1	Absent
5	Sparse
7	Intermediate
9	Dense

5.36 Leaf rolling

The degree of rolling of the leaf blade.

1	Unrolled or very slightly rolled
2	Inrolled (adaxially rolled)
3	Outrolled (abaxially rolled)
4	Spiral

5.37 Flag leaf length

The distance from the collar to the tip of the flag leaf. Unit: cm.

5.38 Flag leaf width

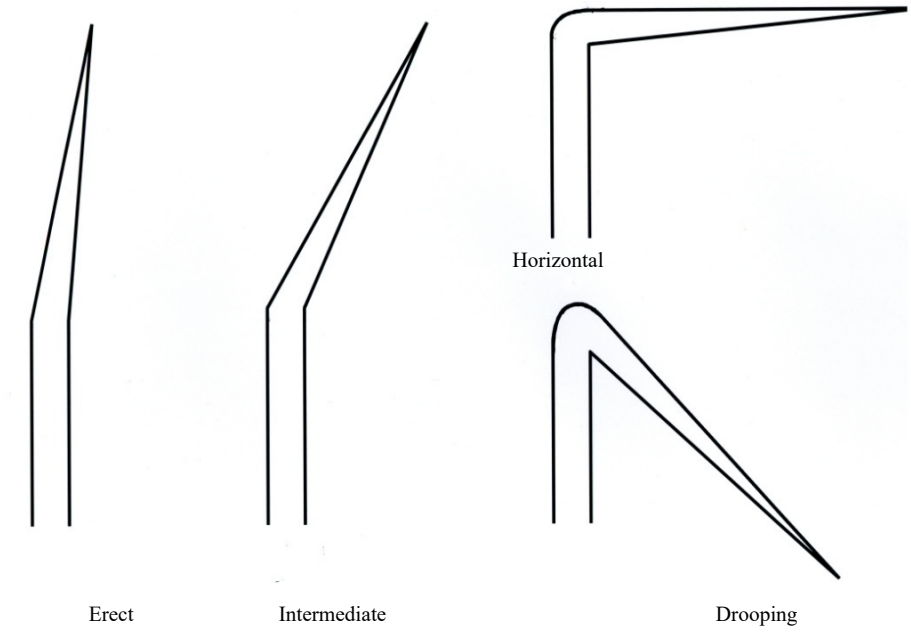
The width of the flag leaf blade at its widest point. Unit: cm.

5.39 Flag leaf angle

The angle between the flag leaf blade and the extension of the culm (Figure 7).

1	Erect
5	Intermediate
7	Horizontal
9	Drooping

Figure 7. Flag leaf angle



5.40 Penultimate leaf length

The distance from the collar to the tip of the penultimate leaf (the second leaf from the top). Unit: cm.

5.41 Penultimate leaf width

The width of the penultimate leaf blade (the second leaf from the top) at its widest point. Unit: cm.

5.42 Penultimate leaf angle

The angle between the line connecting the tip and collar of the penultimate leaf (the second leaf from the top) and the main culm.

1	Erect
5	Horizontal
9	Drooping

5.43 Auricle colour

The colour of the auricles.

1	Colourless
2	Yellow
3	Green
4	Purple

5.44 Ligule colour

The colour of the ligule.

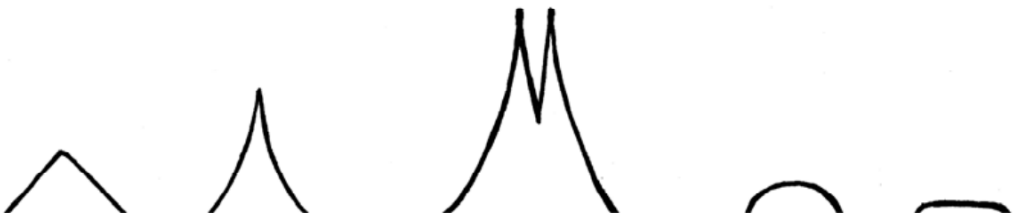
1	Ligule absent
2	White
3	Purple lines
4	Purple

5.45 Ligule shape

The shape of the apex of the ligule (Figure 8).

1	Ligule absent
2	Acute to acuminate
3	Bifid
4	Truncate

Figure 8. Ligule shape



5.46 Leaf collar colour

The surface colour of the leaf collar.

1	Green
2	Purple

5.47 Leaf node colour

The colour of the leaf node.

1	Colourless (white)
2	Green
3	Purple

5.48 Number of leaves on the main culm

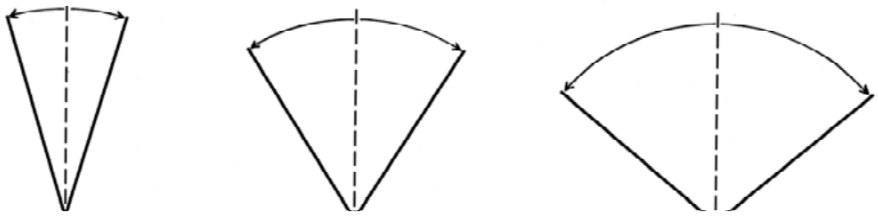
The total number of leaves produced by the main culm during its lifetime.

5.49 Culm angle

The angle between the culm and a line perpendicular to the ground (Figure 9).

1	Erect
3	Intermediate
5	Open
7	Spreading
9	Prostrate

Figure 9. Culm angle



5.50 Culm diameter

The diameter of the culm. Unit: mm.

5.51 Culm node colour

The colour of the culm nodes.

1	Light green
2	Purple

5.52 Culm internode colour

The colour of the culm internodes.

1	Yellow
2	Green
3	Red
4	Purple lines
5	Purple

5.53 Culm node enclosure

Whether the culm nodes are enclosed or exposed.

1	Enclosed
2	Exposed

5.54 Tillering ability

The tillering ability of an individual rice plant.

1	Strong
5	Moderate
9	Weak

5.55 Lodging

The degree of lodging of the culms.

1	Upright
3	Intermediate
5	Leaning
7	Lodged
9	Prostrate

5.56 Stigma colour

The colour of the stigmas.

1	White
2	Yellow
3	Light green
4	Purple

5.57 Single-stigma exsertion rate

Stigma exsertion occurs when a stigma extends completely or partially beyond the hull after flowering. Two forms are distinguished: single-stigma exsertion, in which one of the two stigmas of a spikelet extends beyond the hull; and double-stigma exsertion, in which both stigmas extend beyond the hull.

The single-stigma exsertion rate is the number of spikelets with one exserted stigma as a proportion of the total number of spikelets that have opened. Expressed as a percentage (%).

5.58 Double-stigma exsertion rate

The number of spikelets with both stigmas extending beyond the hull as a proportion of the total number of spikelets that have opened. Expressed as a percentage (%).

5.59 Total stigma exsertion rate

The sum of the numbers of spikelets with single-stigma and double-stigma exsertion as a proportion of the total number of spikelets that have opened. Expressed as a percentage (%).

5.60 Anther shape

The shape of the anthers.

1	Small and rod-shaped
2	Water-soaked appearance
3	Plump

5.61 Anther colour

The colour of the anthers.

1	White or milky white
2	Pale yellow
3	Yellow

5.62 Anther length

The length of the anthers. Unit: mm.

5.63 Spikelet opening angle

The angle between the lemma and palea when the spikelet is fully open.

1	Small
5	Medium
9	Large

5.64 Duration of spikelet opening

The time from the start of separation of the lemma and palea until the spikelet has closed completely. Unit: min.

5.65 Daily flowering duration

During the peak flowering period, the daily interval from the opening of the first spikelet to the closing of the last spikelet. Unit: h.

5.66 Peak flowering time

The time during which 20%-80% of the spikelets flowering on that day are in full bloom.

1	Early
5	Intermediate
9	Late

5.67 Awn length

The length of the awns arising from the tips of the lemmas.

1	Absent
3	Short
5	Medium
7	Long
9	Very long

5.68 Awn colour

The colour of the awns arising from the tips of the lemmas.

1	White
2	Straw yellow
3	Yellow
4	Red
5	Brown
6	Purple
7	Black

5.69 Awn distribution

The distribution of the awns arising from the tips of the lemmas.

1	Absent
3	Very sparse
5	Sparse
7	Intermediate
9	Abundant

5.70 Sterile lemma colour

The colour of the sterile lemmas.

1	Yellow
2	Red
3	Purple

5.71 Sterile lemma length

The length of the sterile lemmas.

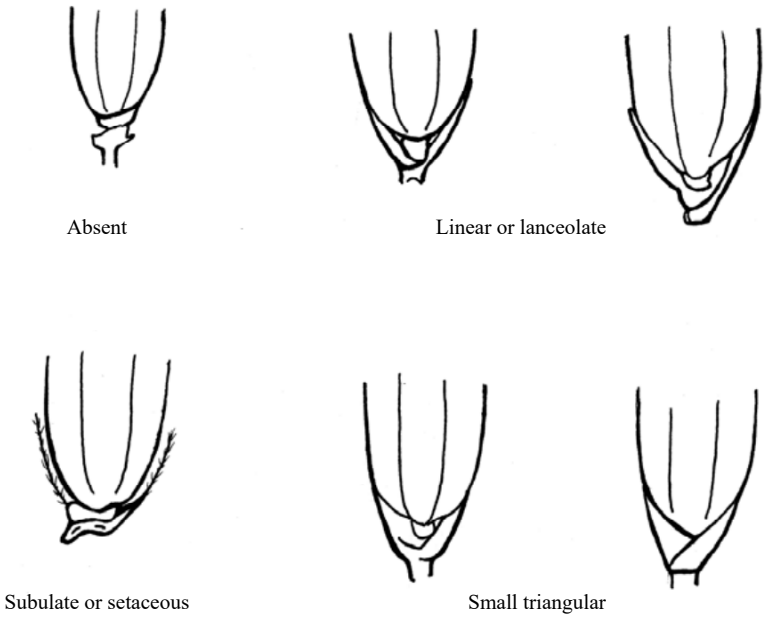
1	Short
3	Medium
5	Long
7	Very long
9	Asymmetrical

5.72 Sterile lemma shape

The shape of the sterile lemmas of mature paddy grains (Figure 10).

1	Absent
2	Linear or lanceolate
3	Subulate or setaceous
4	Small triangular

Figure 10. Sterile lemma shape



5.73 Apiculus colour

The colour of the apiculus.

1	Straw yellow
2	Red
3	Brown
4	Purple
5	Black

5.74 Lemma and palea colour

The colour of the lemma and palea of the paddy grain.

1	Yellow
2	Silver-grey
3	Brown
4	Reddish-brown
5	Purplish-black

5.75 Lemma and palea pubescence

The distribution of hairs on the surfaces of the lemma and palea.

1	Absent
3	Few
5	Intermediate
7	Many
9	Very many

5.76 Grain shattering

The ease with which paddy grains are shed after full maturity.

1	Very low
3	Low
5	Intermediate
7	High
9	Very high

5.77 Male sterility type

The type of male sterility.

1	Cytoplasmic
2	Nuclear
3	Cytoplasmic-nuclear interaction
4	Photoperiod- and temperature-sensitive

5.78 Percentage of sterile plants

The number of plants of a male sterile line that fail to set seed after bagging, as a percentage of the total number of bagged plants. Expressed as a percentage (%).

5.79 Anther dehiscence

The degree of anther dehiscence during flowering.

1	Indehiscent
5	Slight dehiscence
7	Moderate dehiscence
9	Dehiscent

5.80 Degree of pollen sterility

The degree of pollen sterility.

1	Incomplete abortion
2	High degree of abortion
3	Complete abortion

5.81 Pollen abortion type

Classified according to the presence or absence of pollen and its morphology.

1	Pollen-free type
2	Typical abortive type
3	Spherical abortive type
4	Mixed type
5	Partially stained abortive type
6	Stained abortive type

5.82 Restorability of the male sterile line

The ability of F1 hybrids produced by crossing a male sterile line with a typical restorer line to set seed on self-pollination. Evaluated by the seed-setting rate.

1	Weak
3	Relatively weak
5	Relatively strong
7	Strong
9	Very strong

5.83 Maintaining ability of the maintainer line

The ability to maintain the sterility of a male sterile line. Evaluated by the selfed seed-setting rate of the second-generation male sterile line obtained by crossing the maintainer line with the male sterile line.

1	Weak
3	Relatively weak
5	Relatively strong
7	Strong
9	Very strong

5.84 Restoring ability of the restorer line

The ability of F1 hybrids obtained by testcrossing a restorer line with a male sterile line to set seed on self-pollination. Evaluated by the selfed seed-setting rate.

1	Weak
3	Relatively weak
5	Relatively strong
7	Strong
9	Very strong

5.85 Outcrossing seed-setting rate of the male sterile line

The seed-setting rate of a male sterile line resulting from outcrossing after receiving pollen from another plant.

1	Low
3	Relatively low
5	Relatively high
7	High
9	Very high

5.86 Cross-compatibility

The degree of cross-compatibility with typical indica and japonica varieties. Evaluated by the selfed seed-setting rate of the F1 obtained by testcrossing with a tester variety.

1	Poor
5	Intermediate
7	Good
9	Excellent

5.87 Compatibility spectrum

The types and range of germplasm with which a germplasm resource having a given degree of compatibility can cross compatibly.

1	Very narrow
5	Relatively narrow
7	Intermediate
9	Broad

6 Quality characteristics

6.1 Brown rice recovery

The weight of brown rice obtained by dehulling clean paddy as a percentage of the total weight of the paddy grains. Expressed as a percentage (%).

6.2 Milled rice recovery

The weight of milled rice obtained by milling as a percentage of the total weight of the paddy. Expressed as a percentage (%).

6.3 Head rice recovery

The weight of head rice as a percentage of the weight of the paddy grains. Expressed as a percentage (%).

6.4 Head rice kernel length

The maximum distance between the two ends of a head rice kernel. Unit: mm.

6.5 Head rice kernel width

The width of a head rice kernel at its widest point. Unit: mm.

6.6 Head rice kernel length-to-width ratio

The ratio of the length to the width of a head rice kernel.

6.7 Percentage of chalky kernels

The number of rice kernels with chalkiness as a percentage of the total number of kernels in the rice sample. Expressed as a percentage (%).

6.8 Chalky area size

The area of chalkiness within a chalky rice kernel as a percentage of the total area of that kernel. Expressed as a percentage (%).

6.9 Degree of chalkiness

The sum of the chalky areas in the rice kernels as a percentage of the sum of the total areas of the kernels in the sample. Expressed as a percentage (%).

6.10 Translucency

The degree of translucency of head rice kernels, expressed as the relative light transmittance of the rice.

6.11 Aroma

The presence or absence of aroma in rice.

1	Aromatic
2	Non-aromatic

6.12 Gelatinization temperature

The temperature at which rice starch loses its crystallinity during heating and undergoes irreversible swelling to form a starch paste. It is usually determined indirectly using the alkali spreading value. This value measures the extent of disintegration of milled rice kernels in an alkaline solution and indirectly indicates the gelatinization temperature of the rice. A higher alkali spreading value indicates a lower gelatinization temperature.

6.13 Gel consistency

The flow length of the rice gel after alkaline gelatinization of milled rice flour and subsequent cooling. Unit: mm.

6.14 Amylose content

The amylose content of milled rice as a percentage of the dry weight of the milled rice. Expressed as a percentage (%).

6.15 Crude starch content

The crude starch content of milled rice as a percentage of the dry weight of the milled rice. Expressed as a percentage (%).

6.16 Crude protein content

The crude protein content of brown rice or milled rice as a percentage of the dry weight of the brown rice or milled rice. Expressed as a percentage (%).

6.17 Lysine content

The lysine content of brown rice or milled rice as a percentage of the dry weight of the brown rice or milled rice. Expressed as a percentage (%).

6.18 Crude fat content

The crude fat content of brown rice or milled rice as a percentage of the dry weight of the brown rice or milled rice. Expressed as a percentage (%).

7 Abiotic stress resistance

7.1 Seedling-stage drought resistance

The ability of rice to tolerate or withstand drought at the seedling stage. Evaluated using a comprehensive drought resistance coefficient.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.2 Drought resistance over the entire growth period

The ability of rice to tolerate or withstand drought throughout its growth period. Evaluated using the drought resistance index.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.3 Salt tolerance during germination

The ability of rice to tolerate or withstand salt stress during germination. Evaluated using the relative salt injury rate.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.4 Seedling-stage salt tolerance

The ability of rice to tolerate or withstand salt stress at the seedling stage. Evaluated using the mean percentage of dead leaves or the salt injury index.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.5 Alkali tolerance during germination

The ability of rice to tolerate or withstand alkali stress during germination. Evaluated using the germination percentage after alkali stress.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.6 Cold tolerance during germination

The ability of rice to tolerate or withstand low temperatures during germination. Evaluated using the germination percentage at low temperature.

1	Strong
5	Moderate
9	Weak

7.7 Sprout-stage cold tolerance

The ability of rice to tolerate or withstand low temperatures at the sprout stage. Evaluated using seedling mortality following low-temperature stress at the sprout stage.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.8 Seedling-stage cold tolerance

The ability of rice to tolerate or withstand low temperatures at the seedling stage. Evaluated by the degree of reddish-brown withering of the seedlings.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.9 Booting-stage cold tolerance

The ability of rice to tolerate or withstand low temperatures at the booting stage. Evaluated using the percentage of unfilled spikelets.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

7.10 Heat tolerance

The ability of rice plants to tolerate or withstand high temperatures. Evaluated using the percentage of unfilled spikelets after high-temperature stress during flowering.

1	Very strong
3	Strong
5	Moderate
7	Weak
9	Very weak

8 Resistance to diseases and insect pests

8.1 Seedling blast resistance

The resistance of rice at the seedling stage to the rice blast fungus [Magnaporthe grisea (Hebert) Barr nov. comb. (anamorph: Pyricularia oryzae Cav.)].

1	Highly resistant (HR)
2	Resistant (R)
3	Moderately resistant (MR)
4	Moderately susceptible 1 (MS1)
5	Moderately susceptible 2 (MS2)
6	Susceptible 1 (S1)
7	Susceptible 2 (S2)
8	Highly susceptible 1 (HS1)
9	Highly susceptible 2 (HS2)

8.2 Leaf blast resistance

The resistance of rice at the tillering stage to the rice blast fungus [Magnaporthe grisea (Hebert) Barr nov. comb. (anamorph: Pyricularia oryzae Cav.)].

1	Highly resistant (HR)
2	Resistant (R)
3	Moderately resistant (MR)
4	Moderately susceptible 1 (MS1)
5	Moderately susceptible 2 (MS2)
6	Susceptible 1 (S1)
7	Susceptible 2 (S2)
8	Highly susceptible 1 (HS1)
9	Highly susceptible 2 (HS2)

8.3 Panicle neck blast resistance

The resistance of the panicle neck, from heading to maturity, to the rice blast fungus [Magnaporthe grisea (Hebert) Barr nov. comb. (anamorph: Pyricularia oryzae Cav.)].

1	Highly resistant (HR)
2	Resistant (R)
3	Moderately resistant (MR)
5	Moderately susceptible (MS)
7	Susceptible (S)
9	Highly susceptible (HS)

8.4 Panicle node blast resistance

The resistance of the panicle nodes, from heading to maturity, to the rice blast fungus [Magnaporthe grisea (Hebert) Barr nov. comb. (anamorph: Pyricularia oryzae Cav.)].

1	Highly resistant (HR)
2	Resistant (R)
3	Moderately resistant (MR)
5	Moderately susceptible (MS)
7	Susceptible (S)
9	Highly susceptible (HS)

8.5 Bacterial blight resistance

The resistance of rice to the bacterium Xanthomonas oryzae (Uyeda et Ishiyama) Dowson.

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.6 Sheath blight resistance

The resistance of rice to the sheath blight fungus Thanatephorus sasakii (Shirai) Tu and Kimbrough.

0	Immune (IM)
1	Resistant (R)
3	Moderately resistant (MR)
5	Moderately susceptible (MS)
7	Susceptible (S)
9	Highly susceptible (HS)

8.7 Bacterial leaf streak resistance

The resistance of rice to the bacterium Xanthomonas oryzicola Fang et al.

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately susceptible (MS)
7	Susceptible (S)
9	Highly susceptible (HS)

8.8 Whitebacked planthopper resistance

The resistance of rice to the rice insect pest whitebacked planthopper [Sogatella furcifera (Horvath)].

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Moderately susceptible (MS)
9	Susceptible (S)

8.9 Brown planthopper resistance

The resistance of rice to the rice-specialist insect pest brown planthopper [Nilaparvata lugens (Stdl)].

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Moderately susceptible (MS)
9	Susceptible (S)

8.10 Rice gall midge resistance

The resistance of rice to the rice gall midge (Orseolia oryzae Wood et Mason).

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Moderately susceptible (MS)
9	Susceptible (S)

8.11 Striped stem borer resistance

The resistance of rice to the striped stem borer [Chilo suppressalis (Walker)].

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Moderately susceptible (MS)
9	Susceptible (S)

8.12 Yellow stem borer resistance

The resistance of rice to the yellow stem borer [Tryporyza incertulas (Walker)].

0	Immune (IM)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Moderately susceptible (MS)
9	Susceptible (S)

9 Other characteristics

9.1 Uses

The specific uses of rice.

1	Rice for food
2	Rice for health-food uses
3	Rice for food processing
4	Rice for industrial raw materials
5	Rice for feed

9.2 DNA fingerprint profiles and molecular markers

DNA fingerprint profiles of rice germplasm, and the types and characteristic parameters of molecular markers for important agronomic traits.

9.3 Remarks

Detailed explanations of special descriptors or special codes for the rice germplasm.

Translator's notes on the source

2: Reference identifiers, including GB/T 195557.7-2004, NY/T-593-2002 and GB /T 15683-1995, are reproduced as printed in the Chinese source.

5.47: The Chinese source does not further define the anatomical site described as the leaf node in 5.47.

5.83: The expression 'second-generation male sterile line' follows the Chinese source; the source does not explicitly identify this as an F2 generation.

8: Historical scientific names and author citations follow the Chinese source, except for the spelling correction noted for 8.11.

8.11: The species epithet 'supperssalis' in the Chinese source has been corrected to 'suppressalis'.

Figure 9: The supplied Chinese PDF displays three culm-angle sketches. All five category definitions are retained in 5.49; the two additional sketches are not reconstructed.